LECTURE 22: MATH1061/7861
TOPICS: CARDINALITIES
DATE: TUESDAY APRIL 21, 2026

Learning objectives:

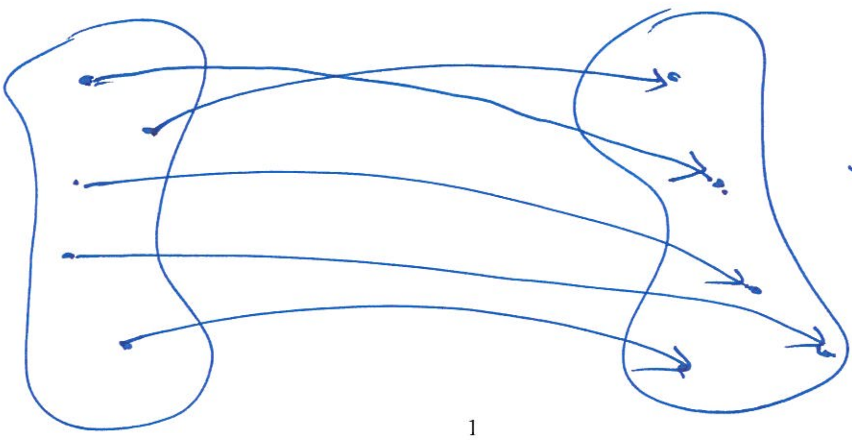
(1) Understand cardinalities of finite sets and infinite sets.

(2) Learn how to prove statements about the cardinalities of sets.

* Reminder: Math 7861 Assignment due this week.

How to measure size of an infinite set?

Note: Suppose we have a bijective function from a set A to a set B and if A and B are finite



then Size of A = Size of B.

We say A and B have the same cardinality if $\exists$ bijective function $A \longrightarrow B$. (We write $|A|=|B|$).

Q2: Prove $|(0,1)| = |\mathbb{R}_{>1}|$.

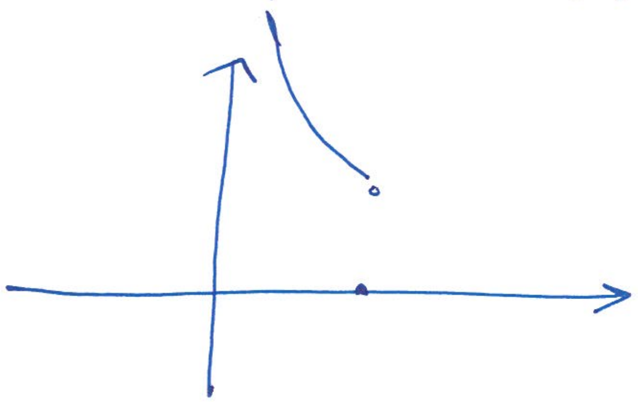
$(0,1) = \{ x \in \mathbb{R} \mid 0 < x < 1 \}$

We need to construct a bijection

$f : (0,1) \longrightarrow \mathbb{R}_{>1}$

Consider $f(x) = \frac{1}{x}$

Observe if $0 < x < 1$ then $\frac{1}{x} > 1$.



So f does map into $\mathbb{R}_{>1}$.

injective: $f(x) = f(y) \Rightarrow \frac{1}{x} = \frac{1}{y} \Rightarrow y = x$ ✓

Surjective: Let $y \in \mathbb{R}_{>1}$. Let $x = \frac{1}{y}$.

Then $x \in (0,1)$. And $f(x) = \frac{1}{x} = \frac{1}{\left(\frac{1}{y}\right)} = y$ ✓

Q1: Prove $|\mathbb{Z}^{\text{odd}}| = |\mathbb{Z}|$.

To do this, we need to construct a bijective function from $\mathbb{Z}$ to $\mathbb{Z}^{odd}$ (or from $\mathbb{Z}^{odd}$ to $\mathbb{Z}$).

Consider
$$f : \mathbb{Z} \longrightarrow \mathbb{Z}^{odd}$$
$$x \longmapsto 2x+1$$

Injective: $f(x) = f(y) \Rightarrow 2x+1 = 2y+1 \Rightarrow x = y$ ✓

Surjective: ~~As~~ The set of odd numbers is defined to be all those numbers of the form $2x+1$.

Alternatively, let $y \in \mathbb{Z}^{odd}$. Then $y = 2k+1$ ✓ for some $k \in \mathbb{Z}$. This means $f(k) = 2k+1 = y$.

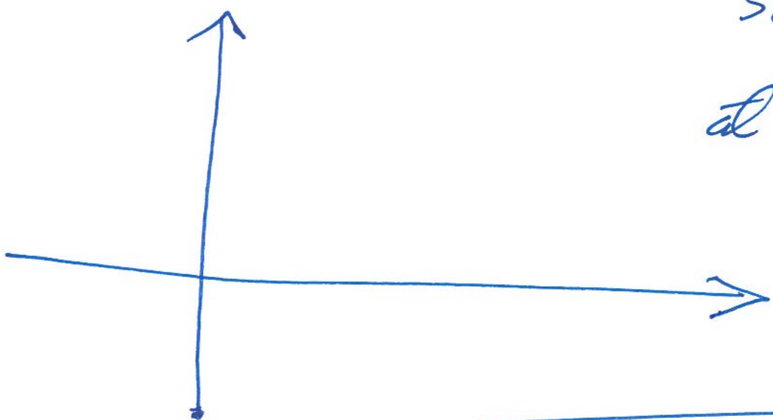
Alternate solution: Consider
$$g : \mathbb{Z}^{odd} \longrightarrow \mathbb{Z}$$
$$x \longmapsto \frac{x-1}{2}$$

Check: This is bijective

<u>Note:</u> $f(x) = \frac{1}{x}$ is not the unique function giving a bijection $(0,1) \longrightarrow \mathbb{R}_{>1}$

e.g. $\tan\left(\frac{\pi}{2} x\right) : (0,1) \longrightarrow \mathbb{R}_{>0}$

$$\tan\left(\frac{\pi}{2} x\right) = \frac{\sin\left(\frac{\pi}{2} x\right)}{\cos\left(\frac{\pi}{2} x\right)}$$



So maybe look at $1 + \tan\left(\frac{\pi}{2} x\right)$

$$\underbrace{x \longmapsto \frac{x}{1-x} + 1}$$

Does this work?